Choosing Image Features for Local Problems: Six Digital Image Processing Cases from Indonesia

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***Abstract*—Many image processing examples in textbooks come from Western settings, such as face databases or car plates in standard formats. This paper looks at six problems that matter in Indonesia and that can be solved with digital image processing: rice leaf disease detection, oil palm fruit ripeness grading, batik motif classification, handwritten Javanese script recognition, road damage detection, and low-light CCTV enhancement. The cases are grouped into three themes: agriculture, cultural heritage, and public infrastructure. For each case we explain the problem, the processing steps used in published work, and which features suit the task, along with the reported results. We find that the best feature depends on one question: *what visual property actually changes between the classes?* Color fits ripeness, texture fits batik and road surfaces, and shape or pixel layout fits handwritten characters.**

***Index Terms*—digital image processing, feature extraction, GLCM, color features, zoning, batik, agriculture, low-light enhancement**

I. Introduction

Indonesia has many problems where a camera and a computer can save time and money. Farmers lose harvests to plant disease, palm oil mills need to sort fruit bunches every day, cultural objects like batik and Javanese script need to be recorded digitally, and roads and public places need to be watched for damage and safety. Most of these jobs are still done by hand, which is slow and depends on the skill of the person doing it.

Digital image processing (DIP) can automate them. A typical system follows five steps: capture the image, clean it (pre-processing), separate the object from the background (segmentation), describe the object with numbers (feature extraction), and make a decision (classification) [1]. This paper focuses on the fourth step, because the choice of features usually decides whether the system works. A classifier can only separate classes if the features show a real difference between them.

Section II briefly lists the feature types used in this paper. Section III presents six cases in three themes. Section IV compares the cases, and Section V concludes.

II. Feature Types Used in This Paper

A. Color Features

Color features are taken from the pixel values of each channel, usually after converting RGB into a space that separates color from brightness, like HSV or L*a*b*. Common choices are the mean, standard deviation, and histogram of each channel. They work when classes differ mainly in color.

B. Texture Features (GLCM)

The Gray-Level Co-occurrence Matrix (GLCM) counts how often a pixel with gray level i appears next to a pixel with gray level j at a given distance and angle [2]. From the normalized matrix $P(i, j)$, several numbers are computed:

$$\text{Contrast} = \sum_{i,j} (i - j)^2 P(i, j) \tag{1}$$

$$\text{Energy} = \sum_{i,j} P(i, j)^2 \tag{2}$$

$$\text{Entropy} = -\sum_{i,j} P(i, j) \log P(i, j) \tag{3}$$

High contrast means sharp changes between neighbors, high energy means a uniform or repeating pattern, and high entropy means a random, complex pattern. Computing GLCM at 0°, 45°, 90°, and 135° captures patterns in different directions.

C. Shape and Layout Features

Shape features describe the outline of a region (area, perimeter, circularity). *Zoning* is a layout feature: a binary character image is divided into an $n \times n$ grid, and the share of black pixels in each cell becomes one feature. A 8×8 grid gives a 64-number vector that describes where the strokes are.

D. Histogram Statistics

The mean, standard deviation, and entropy of the gray-level histogram describe how bright an image is, how much contrast it has, and how much detail it holds. These are useful for measuring image quality rather than for recognizing objects.

III. Cases

Theme 1: Agriculture

A. Rice Leaf Disease Detection

Problem. Rice is Indonesia's main food crop, and diseases such as bacterial leaf blight, brown spot, and leaf smut can spread quickly. Farmers often cannot tell these diseases apart early.

Processing. Azim *et al.* [3] first removed the background using a threshold on the saturation channel, then isolated the

sick area using a threshold on the hue channel. Features were taken only from the sick area, not the whole leaf.

Suitable features. Each disease leaves spots with a different *color* (brown, yellow, dark), a different *shape* (long streaks vs. round spots), and a different surface *texture*. Combining all three gave 86.58% accuracy with an extreme gradient boosting classifier on the UCI rice leaf dataset [3]. The key idea is that cutting out the sick area first makes the features much cleaner.

B. Oil Palm Fruit Ripeness Grading

Problem. Indonesia is the world's biggest palm oil producer. Fresh fruit bunches must be harvested at the right ripeness, because unripe bunches give less oil and overripe bunches raise free fatty acid levels. Graders currently judge ripeness by eye.

Processing. Images of the bunches are captured under controlled lighting, and the fruit region is segmented from the background.

Suitable features. Ripeness shows mainly as a color change, from black or dark purple to reddish orange. So *color features* are the natural choice. Fadilah *et al.* [4] fed color features into an artificial neural network, and found that reducing the features with Principal Component Analysis (PCA) improved accuracy further. Texture and shape add little here, since bunches of all ripeness levels look alike in shape.

Theme 2: Cultural Heritage

C. Batik Motif Classification

Problem. Indonesia has thousands of batik motifs, such as Parang, Kawung, and Megamendung, each tied to a region and meaning. Classifying them automatically helps digital archiving and protects motifs from wrong labeling.

Processing. Images are converted to grayscale and reduced to fewer gray levels (for example 16) so the GLCM stays small.

Suitable features. Batik motifs are defined by their repeating *pattern*, not by color, since the same motif can be made in any color. This makes *texture* the right feature. Minarno *et al.* [5] computed energy, entropy, contrast, and correlation from GLCM at four angles, giving 16 features. With cross-validation, a Support Vector Machine reached 92.3% average accuracy, compared to 78.3% for k-nearest neighbor. Using four angles matters because motifs have different main directions: Parang is built from diagonal lines, while Kawung is built from round shapes, so a single angle would miss part of the pattern.

D. Handwritten Javanese Script Recognition

Problem. Aksara Jawa is a traditional script still taught in schools in Central and East Java. Recognizing handwritten characters can help digitize old manuscripts and support learning apps.

Processing. Each character is converted to a binary image, cropped to its bounding box, and resized to a fixed size so all characters can be compared.

Suitable features. Characters are black strokes on white paper, so color and texture carry almost no information. What matters is *where the strokes are*. Handayani *et al.* [6] used *zoning* features with 64 inputs into a backpropagation neural network and reached 77.00% accuracy. The lower accuracy compared to the other cases shows how hard handwriting is: Javanese characters have many similar shapes, and each person writes differently.

Theme 3: Public Infrastructure and Safety

E. Road Damage Detection

Problem. Potholes and cracks are a real danger, especially for the many motorcycle riders on Indonesian roads. Road agencies need a cheap way to find and classify damage over long distances.

Processing. A camera on a vehicle records the road, and each frame is converted to grayscale before feature extraction.

Suitable features. A smooth road has an even, uniform texture, while damage breaks this pattern. Sartika [7] used GLCM features (dissimilarity, correlation, contrast, energy, and angular second moment) with an SVM to sort road images into five damage types. The overall accuracy was 80%, with F1 scores of 0.95 for potholes and 0.89 for cracks. Potholes are easiest to detect because they create a strong, dark break in the texture.

F. Low-Light CCTV Enhancement

Problem. Many CCTV cameras in Indonesian streets, campuses, and boarding houses (*kos*) are low-cost models. At night their images are dark and noisy, so faces and plates cannot be seen. This case is also the topic of the author's ongoing group project on pixel-based enhancement.

Processing. Pixel-based methods such as gamma correction, histogram equalization, and Contrast Limited Adaptive Histogram Equalization (CLAHE) [8] brighten dark areas. Guo *et al.* [9] review these classic methods alongside newer learning-based ones. Classic methods are fast and need no training data, which suits cheap CCTV hardware.

Suitable features. Unlike the other cases, the features here are not used to recognize an object. They are used to *measure* the image and to *tune* the method. The histogram mean decides how strong the gamma correction should be, the standard deviation shows the contrast, and entropy shows how much detail is visible. After enhancement, metrics like PSNR and SSIM compare the result to a reference image.

IV. Comparison and Discussion

Table I summarizes the six cases. Several lessons stand out.

Pick the feature that matches the change between classes. Ripeness changes color, so color wins. Batik motifs change pattern but not color, so texture wins. Characters change stroke position, so zoning wins. Using a feature that does not follow the real difference, like color for batik, only adds noise.

GLCM is the most reused tool. It appears in batik, road damage, and partly in rice disease. It works for any problem

TABLE I
SUMMARY OF CASES, SUITABLE FEATURES, AND REPORTED RESULTS

Case	Main Feature	Details	Result
Rice leaf disease [3]	Color + shape + texture	Hue/saturation segmentation, features from sick area	86.58% (XGBoost)
Oil palm ripeness [4]	Color	Color features + PCA	PCA improved accuracy (ANN)
Batik motif [5]	Texture	GLCM, 4 stats $\times$ 4 angles	92.3% (SVM)
Aksara Jawa [6]	Layout	Zoning, 64 features	77.00% (ANN)
Road damage [7]	Texture	GLCM, 5 stats	80% (SVM)
Low-light CCTV [9]	Histogram stats	Mean, std, entropy; PSNR/SSIM	Quality, not class

where the surface pattern carries the meaning, and it is cheap to compute.

Segmentation before features helps a lot. In the rice case, taking features only from the sick area instead of the whole leaf gave cleaner features. The same idea applies to palm fruit and road images.

Fewer, better features can beat many features. In the palm oil case, PCA reduced the number of features and still raised accuracy [4].

Finally, the reported accuracies (77–92%) show that hand-crafted features are good but not perfect. Deep learning can learn features on its own, but it needs large labeled datasets, which are rare for local topics like Aksara Jawa or regional batik. For these topics, hand-crafted features remain a practical choice.

V. CONCLUSION

This paper summarized six Indonesian problems that can be solved with digital image processing, across agriculture, cultural heritage, and public infrastructure. For each one, the most suitable feature was the one that follows the real visual difference between classes: color for oil palm ripeness, texture for batik and road damage, a mix of color, shape, and texture for rice disease, stroke layout for Javanese script, and histogram statistics for low-light enhancement. Asking "what actually changes?" before choosing features is a simple rule that leads to simpler and more accurate systems.

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